BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Operating Systems

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. Give a physical security measure and the threat it limits.

2. What is penetration testing?
   A) An attack without permission
   B) Deleting all user accounts
   C) Security testing with permission
   D) Compressing every stored file

3. "POP's main job is sending a new message from a client to a mail server." Correct this. Explain why.

4. A pupil wants matching inboxes and read status on a phone and laptop. "IMAP only sends new outgoing mail." Correct this. Why use IMAP here?

Today's objectives

Explain the core ideas of operating systems using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

User interface

CLI uses typed commands; GUI uses graphics and controls.

Memory management

OS allocates and manages memory for active processes.

Multitasking

OS schedules CPU time among tasks/processes.

File and peripheral management

OS organises files and controls device access.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

User interface
CLI uses typed commands; GUI uses graphics and controls.

Memory management
OS allocates and manages memory for active processes.

Multitasking
OS schedules CPU time among tasks/processes.

File and peripheral management
OS organises files and controls device access.

I Do - Worked model

A laptop plays music while a document is edited. Explain three operating-system responsibilities.

- Allocate CPU processing time to both programs.
- Allocate memory for active processes.
- Manage access to audio hardware and saved document files.
- Describe how scheduling gives an impression of simultaneous progress.

We Do - Guided practice

1. Compare use of a CLI and GUI to manage school computers.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete an assignment showing how much RAM remains after a program loads.

```
freeRAM = 900
usedRAM = 300
remaining = ...
print(remaining)
```

You Do - Independent exam questions

Q2. Name two operating-system functions. [2 marks]

Q3. Explain process scheduling in multitasking. [3 marks]

Q4. Give one benefit of GUI over CLI. [2 marks]

Exit Ticket - True or False?

An OS manages processes, files, memory and devices.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. Lock the server room to limit access and reduce theft or tampering.

2. C - Security testing with permission

3. POP downloads email from a server to a device. IMAP keeps email views in sync across devices.

4. IMAP keeps mail on the server and syncs messages and read status across devices. SMTP sends outgoing mail.

I Do - Explain the model

Allocate CPU processing time to both programs.

Allocate memory for active processes.

Manage access to audio hardware and saved document files.

Describe how scheduling gives an impression of simultaneous progress.

We Do - Model answers

1. Both allow users to interact with OS services.

Code First - Answer

```
freeRAM = 900
usedRAM = 300
remaining = freeRAM - usedRAM
print(remaining)
```

Expected output: 600

You Do - Self-mark

Q2: Memory management, process scheduling, file management, user interface or peripherals (any two).

Q3: The OS allocates CPU time to processes so multiple programs can make progress.

Q4: Many users find labelled visual controls easier to learn/use.

GREEN PEN: Choose the hardest answer. Correct it on lined paper.